

AETHER

Operations Portal - Technical Documentation

Comprehensive CRM, Inventory Control & Sales Challan Ledger System

Aether Operations Portal - Technical & API Documentation

Aether is a role-based Enterprise Operations Portal integrating Customer Relationship Management (CRM), Real-time Inventory Control, and Commercial Sales Challan Workflows.

?? System Architecture

The following diagram illustrates the data flow and architectural design of the Aether portal:

graph TD

```
subgraph Frontend (Pure CSS + React)
```

?? Database Entity-Relationship Diagram (ERD)

Aether enforces snapshot integrity for historical transactions. Below is the relational database layout:

```
erDiagram
```

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? REST API Endpoint Reference

All endpoints (except Authentication) require a Bearer JWT Token passed in the `Authorization` header:

`Authorization: Bearer <your\_jwt\_token>`

? Authentication Services

- **POST /api/auth/login**

* **Payload:**

```
```json
{ "username": "admin", "password": "admin123" }
```
```

* **Response:**

```
```json
{
 "token": "eyJhbGciOi...",
 "user": { "id": "1", "username": "admin", "name": "System Administrator", "role": "Admin" }
}
```

### ? Operations Dashboard

- **GET /api/dashboard/stats**

### ? Customer Relationship Management (CRM)

- **GET /api/customers**

\* **Description:** Retrieves a paginated list of accounts.

- **POST /api/customers**

- **PUT /api/customers/:id**

- **POST /api/customers/:id/notes**

### ? Inventory Control

- **GET /api/products**

\* **Description:** Retrieves active product catalog.

- **POST /api/products**

- **PUT /api/products/:id**

- **GET /api/products/:id/logs**

? Commercial Sales Challans

- \*\*`GET /api/challans`\*\*  
  
\* \*Description\*: Retrieves generated challans.
- \*\*`POST /api/challans`\*\*
- \*\*`PATCH /api/challans/:id/status`\*\*

? Key Architectural Design Decisions

1. Snapshot Integrity Design Pattern

A common failure point in ERP systems is when a product's price or customer's address is updated, causing older invoices/challans to change value retroactively.

- \*\*Solution\*\*: When a sales challan is confirmed, the system serializes a snapshot of the `Customer` and `Product` details as a frozen JSON string in the `customerSnapshot` and `productSnapshot` fields.

2. Double-Transaction Stock Lock

- \*\*Draft Status\*\*: Challans saved as `Draft` reserve zero inventory, allowing operators to build quotes without blocking real stock.

3. Glassmorphic Notification Toast System

Instead of browser alerts, a custom React context tracks alerts, applies slide-in keyframe animations, and clears them automatically after 4 seconds to maintain a premium application flow.